

Checking for Further Improvements

If you look closely at [Figure 6-2](#) or [Figure 6-3](#), you might be able to imagine some specific improvements that could be made. You could even attempt those improvements yourself and check whether they worked by using our `howfar()` function. For example, maybe you look at our initial random itinerary:

```
initial_itinerary =  
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26,  
\27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39]
```

and you think you could improve the itinerary by switching the order of the salesman's visits to city 6 and city 30. You can switch them by defining this new itinerary with the numbers in question switched (shown in bold):

```
new_itinerary =  
[0, 1, 2, 3, 4, 5, 30, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27,  
\28, 29, 6, 31, 32, 33, 34, 35, 36, 37, 38, 39]
```

We can then do a simple comparison to check whether the switch we performed has decreased the total distance:

```
print(howfar(genlines(cities, initial_itinerary)))  
print(howfar(genlines(cities, new_itinerary)))
```

If the `new_itinerary` is better than the `initial_itinerary`, we might want to throw out the `initial_itinerary` and keep the new one. In this case, we find that the new itinerary has a total distance of about 16.79, a very slight improvement on our initial itinerary. After finding one small improvement, we can run the same process again: pick two cities, exchange their locations in the itinerary, and check whether the distance has decreased. We can continue this process indefinitely, and at each step expect a reasonable chance that we can find a way to decrease the traveling distance. After repeating this process many times, we can (we hope) obtain an itinerary with a very low total distance.

It's simple enough to write a function that can perform this switch-and-check process automatically ([Listing 6-3](#)):